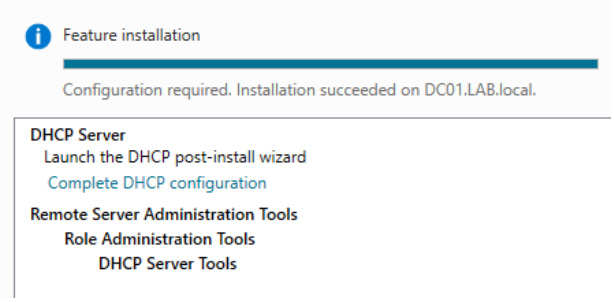
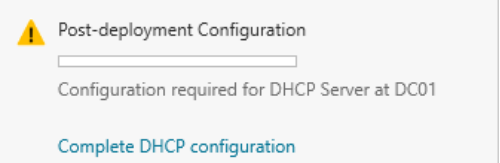
DHCP LAB via Local AD

1. Installing the DHCP Server on DC01

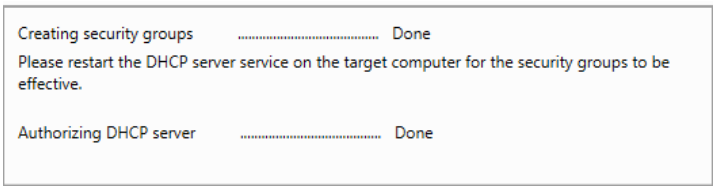
Server Manager -> Manage -> Add Roles and Features -> Role-based or feature-based installation -> DC01 highlighted -> Next -> DHCP Server box checked -> Add Features -> Next all the way -> Install

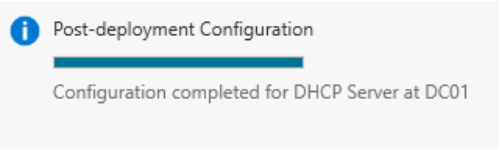


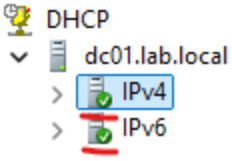
2. Authorizing the DHCP Server in AD



to

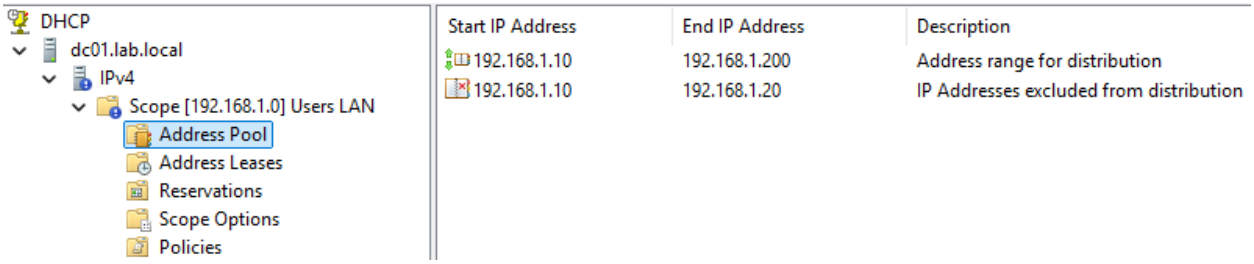


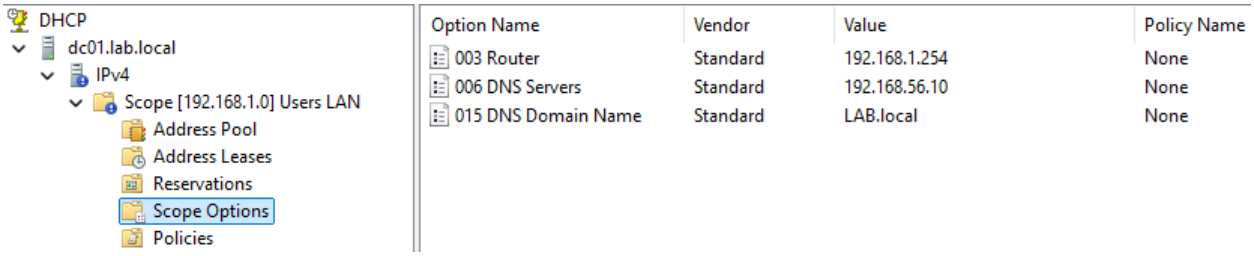




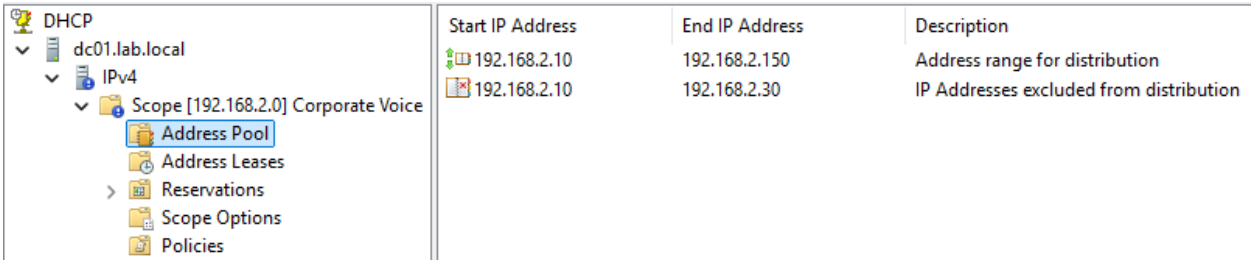
I will now build DHCP scopes. I will create three – Users LAN, Corporate Voice, and Guest Wi-Fi. These will represent three different network segments in a real corporate-like environment.

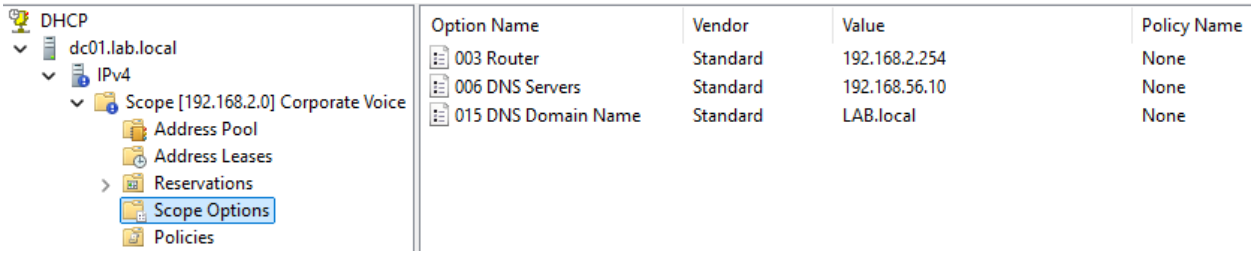
3. Create Scope 1 — Users LAN



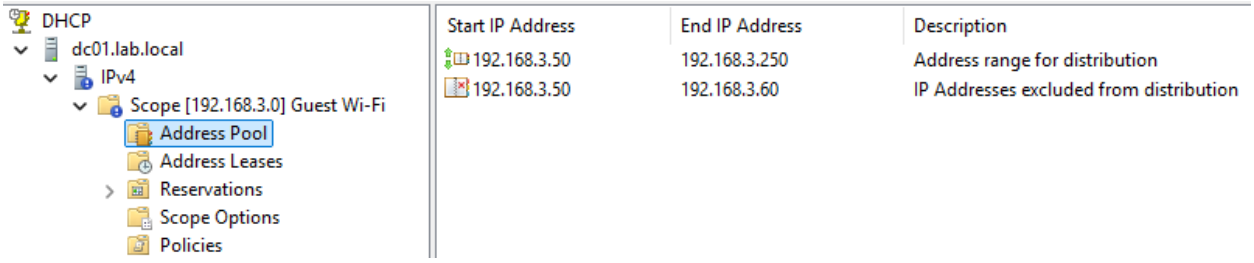


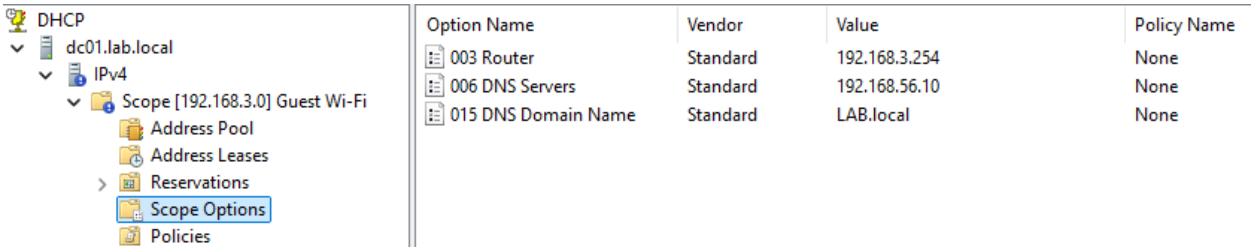
4. Create Scope 2 – Corporate Voice





5. Create Scope 3 – Guest Wi-Fi

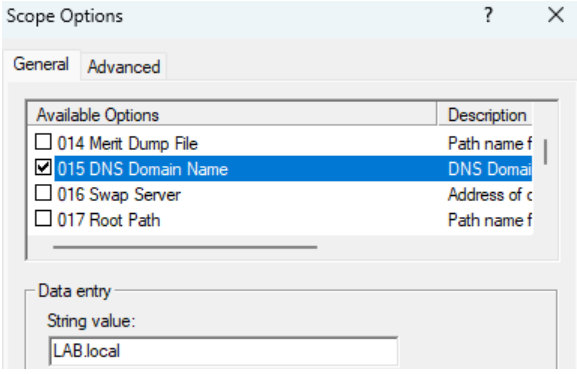




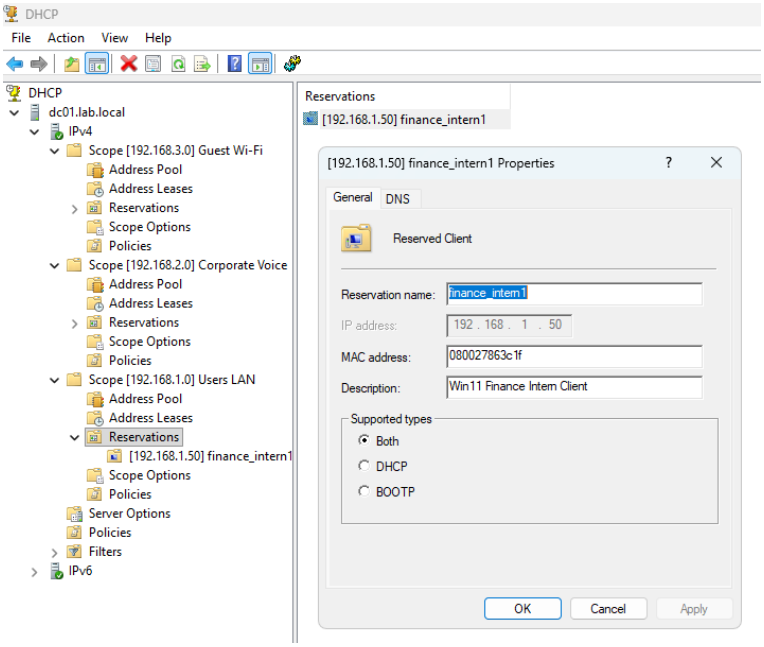
All done:

6. Adding DNS Domain Name Option to All Three Scopes

All three scopes have them checked under scope -> Scope Options -> Configure Options as so:

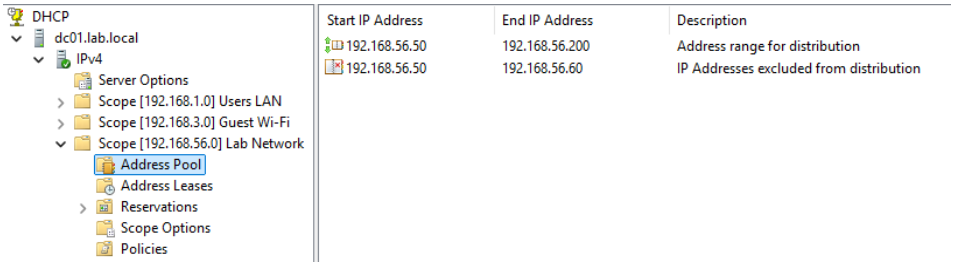


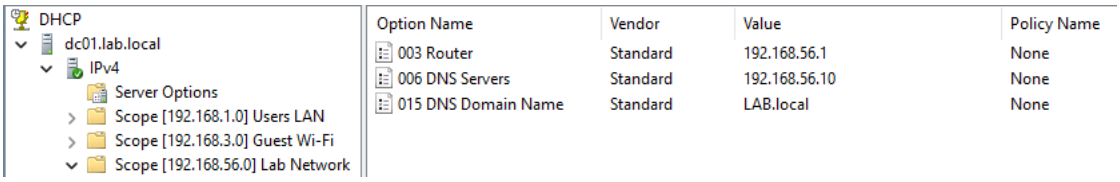
7. Creating a DHCP Reservation for the W11 Client (via Users LAN scope)



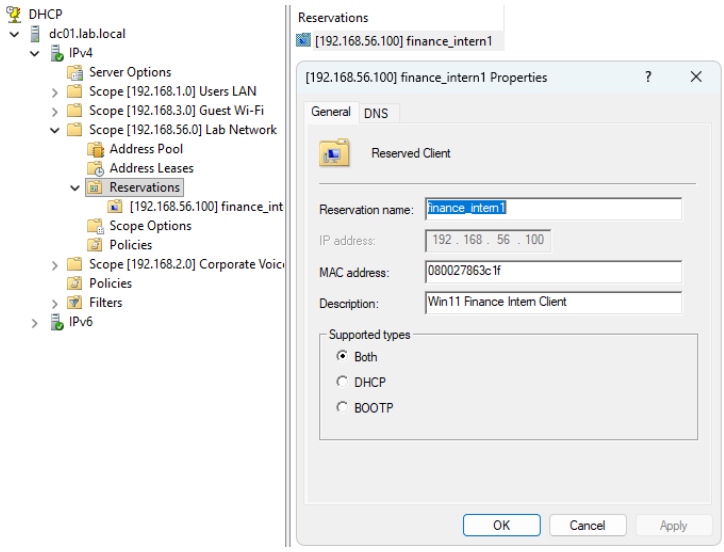
8. Live Lease Test – Win11 Client

So, the three scopes created are correctly configured but they represent networks that don't physically exist in my lab. In a real corporate environment, those scopes would work because there would be a router or Layer 2 switch with a DHCP helper address configured to relay requests from those subnets to the DHCP server. Without the relay, a DHCP server only responds to requests that arrive on a subnet it has a matching scope for. My lab network where the VMs live runs on 192.168.56.x so I need a scope here for real lease testing:





Switching the reservation for “finance_intern1” over to the “Lab Network” scope:



Everything worked properly. All the fields check out:

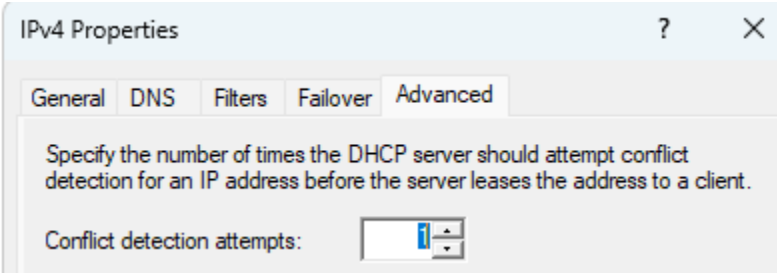
- IP: 192.168.56.100 — reservation working
- DHCP Server: 192.168.56.10 — DC01 is handing out the lease
- Default Gateway: 192.168.56.1
- DNS Server: 192.168.56.10 — pointing to the domain controller
- DNS Suffix: LAB.local
- Lease duration: 8 hours
- DHCP Enabled: Yes

```
C:\Users\[redacted]>ipconfig /all

Windows IP Configuration

   Host Name . . . . . . . . . . . . : finance_user1
   Primary Dns Suffix  . . . . . . . : LAB.local
   Node Type . . . . . . . . . . . . : Hybrid
   IP Routing Enabled. . . . . . . . : No
   WINS Proxy Enabled. . . . . . . . : No
   DNS Suffix Search List. . . . . . : LAB.local

Ethernet adapter Ethernet:

   Connection-specific DNS Suffix  . : LAB.local
   Description . . . . . . . . . . . : Intel(R) PRO/1000 MT Desktop Adapter
   Physical Address. . . . . . . . . : 08-00-27-86-3C-1F
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes
   Link-local IPv6 Address . . . . . : fe80::b5fb:ded1:a682:1187%4(Preferred)
   IPv4 Address. . . . . . . . . . . : 192.168.56.100(Preferred)
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Lease Obtained. . . . . . . . . . : Monday, April 13, 2026 2:50:54 PM
   Lease Expires . . . . . . . . . . : Monday, April 13, 2026 10:50:52 PM
   Default Gateway . . . . . . . . . : 192.168.56.1
   DHCP Server . . . . . . . . . . . : 192.168.56.10
   DHCPv6 IAID . . . . . . . . . . . : 101187623
   DHCPv6 Client DUID. . . . . . . . : 00-01-01-00-31-67-9C-B5-08-00-27-86-3C-1F
   DNS Servers . . . . . . . . . . . : 192.168.56.10
   NetBIOS over Tcpip. . . . . . . . : Enabled
```

One last verification step:

```
PS C:\WINDOWS\system32> Get-DhcpServerv4Lease -ScopeId 192.168.56.0

IPAddress       ScopeId         ClientId            HostName            AddressState        LeaseExpiryTime
---------       -------         --------            --------            ------------        ---------------
192.168.56.100  192.168.56.0    08-00-27-86-3c-1f   finance_user1.LAB... ActiveReservation
```

Finance_intern1 has an Active state. All is well.

Conflict detection

This makes the server ping an IP before offering it to make sure nothing is already using it statically.